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NETWORK & INFORMATION SECURITY

# OpenVAS

## *Open Vulnerability Assessment System*

*Comprehensive Network Security Scanning & Vulnerability Management*

**Submitted By**

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# 1.Abstract

OpenVAS (Open Vulnerability Assessment System) is a comprehensive, open-source vulnerability scanning and management framework maintained by Greenbone Networks as part of the Greenbone Vulnerability Management (GVM) suite. Originally forked from the Nessus 2.x codebase in 2006, following Tenable's decision to make Nessus proprietary, OpenVAS has evolved into a mature, enterprise-capable security platform licensed under the GNU GPL. It provides organisations and security professionals with a free yet powerful means to identify, classify, and remediate vulnerabilities across their network infrastructure.

This report presents a technical examination of OpenVAS, covering its multi-component architecture, end-to-end scanning workflow, and the intelligence engine that powers it — the Network Vulnerability Tests (NVTs). The report includes a practical lab experiment against a Metasploitable target, a detailed analysis of findings, and an evaluation of real-world applicability and limitations. The findings confirm that OpenVAS is an exceptionally capable, cost-effective vulnerability management solution ideally suited for network auditing, penetration testing, compliance assurance, and educational environments.

# 2. Objectives

This report aims to achieve the following objectives:

- To examine the architecture, components, and scanning workflow of OpenVAS/GVM, with a focus on its NVT-based detection intelligence and CVSS-based severity classification.
- To demonstrate the complete installation and configuration of OpenVAS in a controlled lab environment using Kali Linux and Metasploitable 2 on an isolated VirtualBox network.
- To conduct a full vulnerability scan against an intentionally vulnerable target and analyse the findings in terms of detection coverage, CVSS scoring accuracy, and false positive behaviour.
- To evaluate the practical applicability of OpenVAS for real-world security use cases — including network auditing, penetration testing, and regulatory compliance — alongside its known limitations.

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# 3. Introduction

## 3.1 What is OpenVAS?

OpenVAS is a full-featured, open-source vulnerability scanner and management solution maintained by Greenbone Networks as a central component of their Greenbone Vulnerability Management (GVM) framework. Unlike many security tools that address only a narrow slice of the vulnerability management lifecycle, OpenVAS handles every stage — from scan configuration and automated execution, through result scoring and analysis, to the generation of comprehensive remediation reports.

It supports both authenticated scanning (where the scanner logs in with credentials to perform deep internal checks) and unauthenticated scanning (where it probes services externally as an attacker would). The community edition provides access to more than 50,000 Network Vulnerability Tests (NVTs) through the Greenbone Community Feed, while the Enterprise Feed extends coverage to over 90,000 NVTs, regularly updated for Linux, Windows, network devices, web applications, and databases.

## 3.2 Historical Background

OpenVAS originated from the 2006 community fork of Nessus 2.x, following Tenable Network Security's decision to close the Nessus source code. Greenbone Networks assumed stewardship in 2008, and by 2017 formally rebranded the platform as the Greenbone Vulnerability Management (GVM) framework, reflecting its evolution into a full vulnerability management lifecycle solution.

| Year | Milestone |
|---|---|
| **2005** | Tenable makes Nessus proprietary, triggering a community response. |
| **2006** | Community forks Nessus 2.x; OpenVAS is born as a fully open-source vulnerability scanner. |
| **2008** | Greenbone Networks assumes stewardship, bringing professional development and commercial feed support. |
| **2017** | Platform rebranded as Greenbone Vulnerability Management (GVM), reflecting full lifecycle scope. |
| **2019+** | GVM 11+ introduces modular architecture: OpenVAS Scanner, gvmd, GSA, and OSPd components. |

### 3.3 Importance in Cybersecurity

The modern threat landscape is characterised by a relentless rate of vulnerability disclosure. Organisations that fail to proactively identify and remediate weaknesses are at significant risk of data breaches, ransomware attacks, and regulatory penalties. OpenVAS enables security teams to perform network audits, identify unpatched systems and open ports, enumerate misconfigurations, and demonstrate compliance with regulatory frameworks such as PCI-DSS, HIPAA, and ISO 27001. Its CVSS-based scoring ensures that findings are prioritised by severity, allowing efficient allocation of remediation resources.

### 3.4 Open-Source Nature and Relevance

Released under the GNU General Public License (GPL), OpenVAS may be freely used, modified, and distributed. This has cultivated a large, active community of contributors who continuously improve the codebase and add new NVTs. For organisations operating under tight budgets — startups, academic institutions, healthcare providers, and non-profits — OpenVAS delivers professional-grade scanning without the $3,000+ licensing costs of commercial alternatives. Unlike black-box commercial scanners, OpenVAS allows administrators to inspect exactly what each NVT is testing and integrate it into broader workflows through the GMP API.

## 4. Features of the Tool

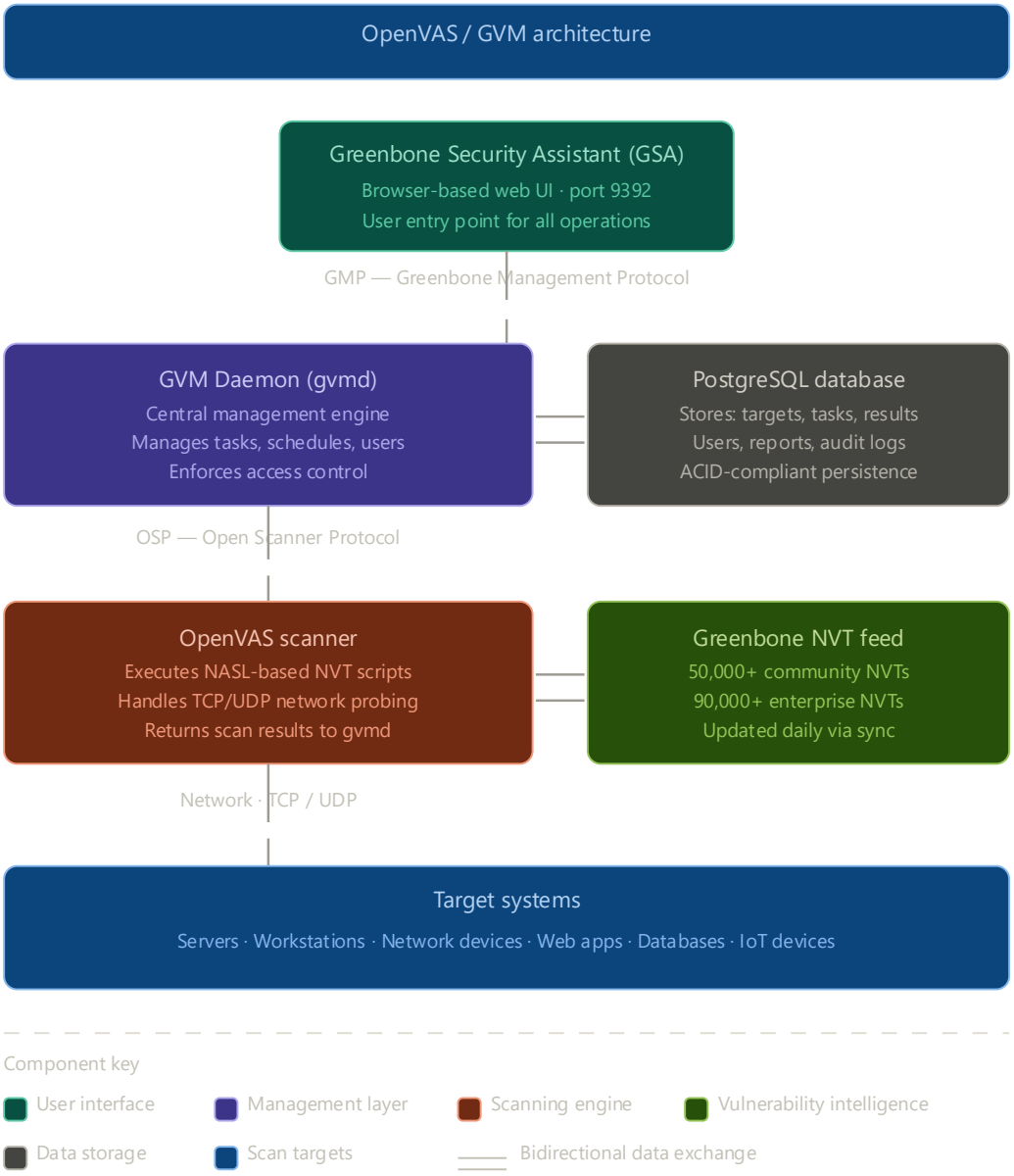
### 4.1 Authenticated vs. Unauthenticated Scanning

An unauthenticated scan probes a target externally, detecting open ports, service banners, and externally exploitable vulnerabilities. Authenticated scanning goes further — logging into target systems using SSH key pairs, Windows domain credentials, or SNMP community strings — to enumerate installed software, running service versions, configuration files, and user account policies. This provides dramatically deeper vulnerability intelligence and is particularly important in enterprise environments where internal misconfigurations represent the majority of real risk.

### 4.2 NVT Feed — Detection Intelligence

The Network Vulnerability Test (NVT) feed is the backbone of OpenVAS's detection capability. Each NVT is a self-contained script written in NASL (Nessus Attack Scripting Language) encoding the detection logic for a specific vulnerability, with embedded CVE references, CVSS scores, and remediation guidance. NVTs are organised into logical families: Web Applications, Linux/Unix, Windows, Databases, Network Infrastructure, and SCADA Systems. The Greenbone Community Feed provides 50,000+ NVTs free of charge, updated daily; the Enterprise Feed extends this to 90,000+.

## 4.3 System Architecture




## 4.4 CVSS Severity Classification

OpenVAS uses the Common Vulnerability Scoring System (CVSS) as its primary severity framework. Scores are computed from a vector string encoding Base Score, Temporal Score, and Environmental Score. Findings are classified into five severity bands:

| Severity | CVSS Score | Risk Level | Required Action |
|---|---|---|---|
| Critical | 9.0 – 10.0 | Immediately Exploitable | Emergency patch. Remotely exploitable, no authentication required. Fix within 24–48 hrs. |
| High | 7.0 – 8.9 | Serious Risk | Urgent remediation. Exploitable with low complexity. Patch within one week. |
| Medium | 4.0 – 6.9 | Moderate Risk | Scheduled patching. Requires user interaction or local access. |
| Low | 0.1 – 3.9 | Minor Risk | Low priority. Limited impact. Monitor and remediate when convenient. |
| Log/Info | 0.0 | Informational | No immediate action needed. Useful for configuration auditing. |

## 4.5 Flexible Scheduling and Continuous Monitoring

OpenVAS supports configurable scan scheduling — hourly, daily, weekly, or monthly — transforming it from a point-in-time tool into a continuous monitoring solution. Security teams can automate weekly infrastructure scans, daily critical-asset checks, and monthly compliance audits without manual intervention.

## 4.6 Multi-Format Reporting

Scan results can be exported as PDF (executive briefings and compliance documentation), XML (SIEM and ticketing system integration), HTML (web-based stakeholder sharing), and CSV (spreadsheet analysis). Each format contains the complete set of findings with CVE identifiers, CVSS scores, affected hosts/ports, detailed descriptions, and remediation recommendations.

## 4.7 Policy and Compliance Scanning

Built-in scan configurations align with PCI-DSS, HIPAA, and CIS Benchmarks. PCI-DSS Requirement 11.2 mandates quarterly vulnerability scans — a requirement OpenVAS fulfils at zero licensing cost, with CVE-linked reports that provide auditors with systematic evidence of due diligence.

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## 4.8 GMP API for Automation and Integration

The Greenbone Management Protocol (GMP) is an XML-based API enabling programmatic creation of targets, launch of scans, retrieval of results, and integration into CI/CD pipelines, SIEM platforms, and custom dashboards. Official Python bindings (python-gvm) are maintained by Greenbone, reducing the barrier to automation.

## 4.9 Comparison with Other Vulnerability Scanners

OpenVAS competes directly with leading commercial vulnerability scanners. The table below summarises key differences:

| Feature | OpenVAS (GVM) | Nessus Professional | Qualys VMDR |
|---|---|---|---|
| **Cost** | Free (Open Source / GNU GPL) | ~$3,000+ / year | SaaS subscription |
| **NVT / Plugins** | 50,000+ Community; 90,000+ Enterprise | 59,000+ | 80,000+ |
| **Cloud Scanning** | No (self-hosted) | No (self-hosted) | Yes (cloud-native) |
| **Setup** | Moderate (Linux expertise) | Easy (wizard-driven) | Very Easy (no install) |
| **Auth. Scanning** | Yes (SSH, WMI, SNMP) | Yes (extensive) | Yes (extensive) |
| **Compliance** | PCI-DSS, HIPAA (basic) | PCI, HIPAA, CIS (full) | PCI, HIPAA, SOC2 |
| **API Support** | Yes (GMP / python-gvm) | Yes (REST API) | Yes (REST API) |
| **Source Code** | Fully open (auditable) | Proprietary | Proprietary |

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# 5. Installation & Setup

OpenVAS was set up using Oracle VirtualBox as the virtualization platform. Two virtual machines were created: one running Kali Linux (which hosts OpenVAS) and one running Metasploitable (the intentionally vulnerable scan target). Both VMs were placed on the same Host-Only network adapter to allow direct communication between them while remaining isolated from the external network.

Lab Environment:
Hypervisor: Oracle VirtualBox
Scanner Host: Kali Linux (min. 4 GB RAM, 30 GB disk)
Scan Target: Metasploitable 2
Network Mode: Host-Only Adapter (isolated internal network)

## 5.1 Step-by-Step Installation

The following steps describe the complete process to install and configure OpenVAS within the VirtualBox lab environment.

**Step 1 —** Install Oracle VirtualBox — Download and install Oracle VirtualBox from https://www.virtualbox.org. Once installed, open VirtualBox Manager.

**Step 2 —** Set up the Kali Linux VM — Download the Kali Linux ISO or a pre-built VirtualBox image from https://www.kali.org/get-kali. Create a new VM in VirtualBox, allocate at least 4 GB of RAM and 30 GB of disk space, and attach the Kali image. Boot into Kali Linux and complete the initial setup.

**Step 3 —** Set up the Metasploitable VM — Download the Metasploitable 2 VirtualBox image from https://sourceforge.net/projects/metasploitable. Import it into VirtualBox. Set both VMs to use the same network adapter type (Host-Only Adapter) so they can communicate on the same isolated subnet.

**Step 4 —** Update the Kali system — Inside the Kali Linux VM, open a terminal and run the system update commands to ensure all packages are current before installation.

```
sudo apt update && sudo apt upgrade -y
```

**Step 5 —** Install OpenVAS — Install the full Greenbone Vulnerability Management suite from the Kali Linux repositories using the following command:

```
sudo apt install openvas -y
```

**Step 6 —** Run the GVM setup script — Execute the setup script which initializes the PostgreSQL database, generates TLS certificates, and downloads the initial NVT feed. This step takes several minutes as it downloads the full NVT library.

Once complete, the admin password is displayed in the terminal — note it down carefully.

```
sudo gvm-setup
```

**Step 7 —** Verify the installation — Run the check script to confirm all GVM components are properly configured and services are healthy:

```
sudo gvm-check-setup
```

**Step 8 —** Start GVM services — Start all required GVM daemon services:

```
sudo gvm-start
```

**Step 9 —** Access the Web Interface — Open a browser inside the Kali VM and navigate to the GSA web interface. Accept the self-signed certificate warning and log in using the admin credentials generated in Step 6.

```
https://127.0.0.1:9392
```

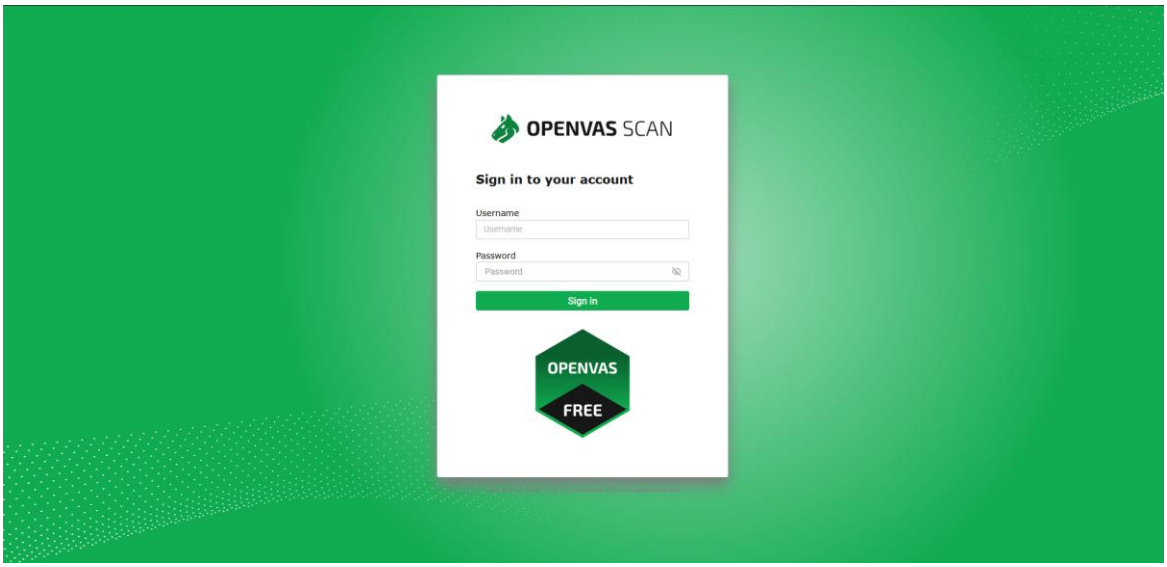


Figure 5.1 — Greenbone Security Assistant (GSA) web interface after successful login

**Step 10 —** Verify the NVT Feed — After logging in, navigate to Administration > Feed Status and confirm the NVT feed is synchronized and up to date. The feed status page shows the last sync timestamp and total NVT count.

# 6. Experimentation & Results

To validate OpenVAS in a practical security assessment scenario, a full vulnerability scan was performed against Metasploitable 2 — an intentionally vulnerable Linux virtual machine widely used in security training and research. Both machines were hosted on an isolated VirtualBox Host-Only network to allow full network communication while containing all traffic within the lab environment.

## 6.1 Target Setup

Metasploitable 2 was run as a virtual machine within the same VirtualBox Host-Only network as the Kali Linux scanner. The IP address of the Metasploitable instance on the local network was identified as 192.168.29.97. This IP was used as the designated scan target in OpenVAS. Metasploitable intentionally runs a wide range of outdated, misconfigured, and deliberately vulnerable services across multiple ports, making it an ideal controlled target for demonstrating OpenVAS's detection capabilities.

## 6.2 Creating the Scan Task

A new scan task was created using the Task Wizard accessible from the top-left menu of the GSA dashboard. The IP address of the Metasploitable virtual machine (192.168.29.97) was entered in the hostname field. The default Full and Fast scan configuration was selected, which runs all applicable NVTs against all detected services on the target host.

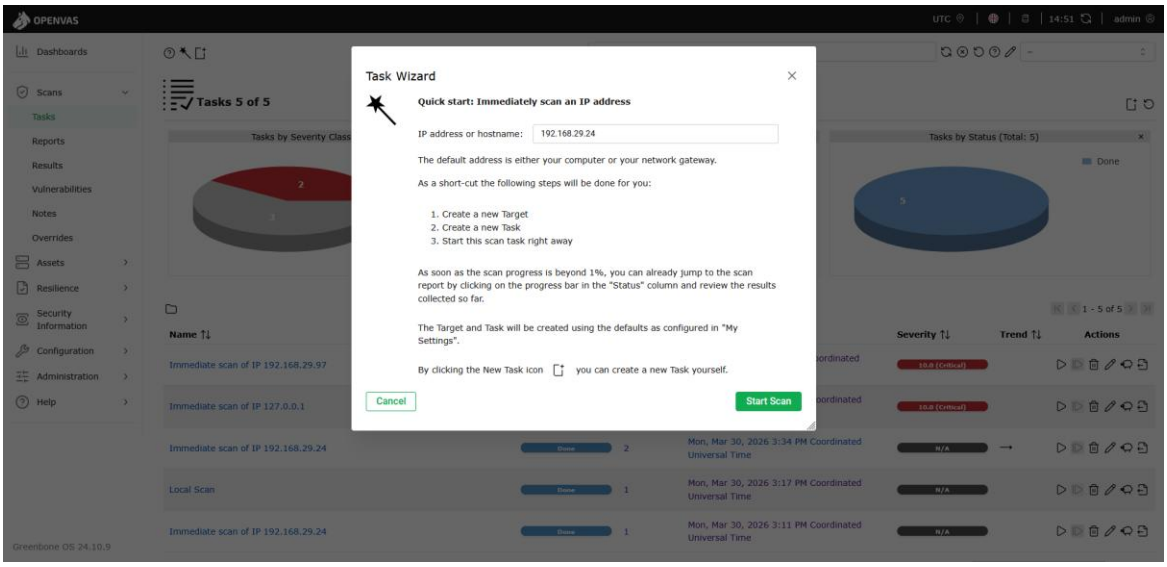


Figure 6.1 — OpenVAS Task Wizard with Metasploitable target IP configured

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## 6.3 Running the Scan

The scan was launched by clicking the Start button on the created task. The scan ran for approximately 30 to 60 minutes, during which OpenVAS probed the Metasploitable machine across all ports and services. The GSA dashboard provided real-time progress updates throughout the scan execution, showing the percentage of NVTs completed and the number of hosts processed.

## 6.4 Scan Results Summary

Upon completion, the scan report was opened in the GSA interface. The scan identified a total of 630 findings across 19 ports, referencing 34 CVEs. The findings were distributed across the Critical, Medium, Low, and Informational severity bands as follows:

| Severity | Key Findings | Count / Detail |
|---|---|---|
| **Critical** | dRuby/DRb Multiple RCE, Apache Tomcat AJP Ghostcat RCE, vsftpd Backdoor, DistCC RCE (CVE-2004-2687), TWiki XSS/Command Execution, Ingreslock Backdoor, MySQL Default Credentials, rlogin Passwordless Login, rexec open service, OS End of Life | Multiple findings — immediate exploitation risk |
| **Medium** | Weak service configurations, outdated software version banners, insecure protocol exposure across multiple open ports | Various ports across 19 detected services |
| **Low / Info** | Service version disclosures, unnecessary open ports, default configuration issues | 630 total findings, 34 CVEs referenced |

## 6.5 Detailed Critical Vulnerabilities Identified

The following critical vulnerabilities were detected on the target host (192.168.29.97):

- **Distributed Ruby (dRuby/DRb) Multiple RCE:** Remote code execution vulnerabilities in the dRuby service, allowing unauthenticated attackers to execute arbitrary commands.
- **Apache Tomcat AJP RCE — Ghostcat (CVE-2020-1938):** Critical deserialization vulnerability in the AJP connector on port 8009 allowing remote code execution without authentication.

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- **vsftpd Compromised Source Packages Backdoor:** A backdoor deliberately introduced into vsftpd 2.3.4, listening on port 6200/tcp and providing instant root shell access.
- **DistCC RCE (CVE-2004-2687):** Remote code execution via the DistCC distributed compiler service on port 3632/tcp — a classic Metasploitable exploit target.
- **TWiki Multiple XSS / Command Execution:** Multiple vulnerabilities in the TWiki web application allowing cross-site scripting and OS command execution.
- **Possible Backdoor: Ingreslock:** A suspicious service listening on port 1524/tcp providing unauthenticated root shell — a deliberate backdoor in Metasploitable.
- **MySQL / MariaDB Default Credentials:** Database server accessible with default empty root password, rated CVSS 9.8.
- **rlogin Passwordless Login:** The rlogin service configured to allow login without any password authentication — CVSS 10.0.
- **Operating System End of Life (EOL):** The underlying OS was flagged as End of Life, meaning no further security patches are available from the vendor.

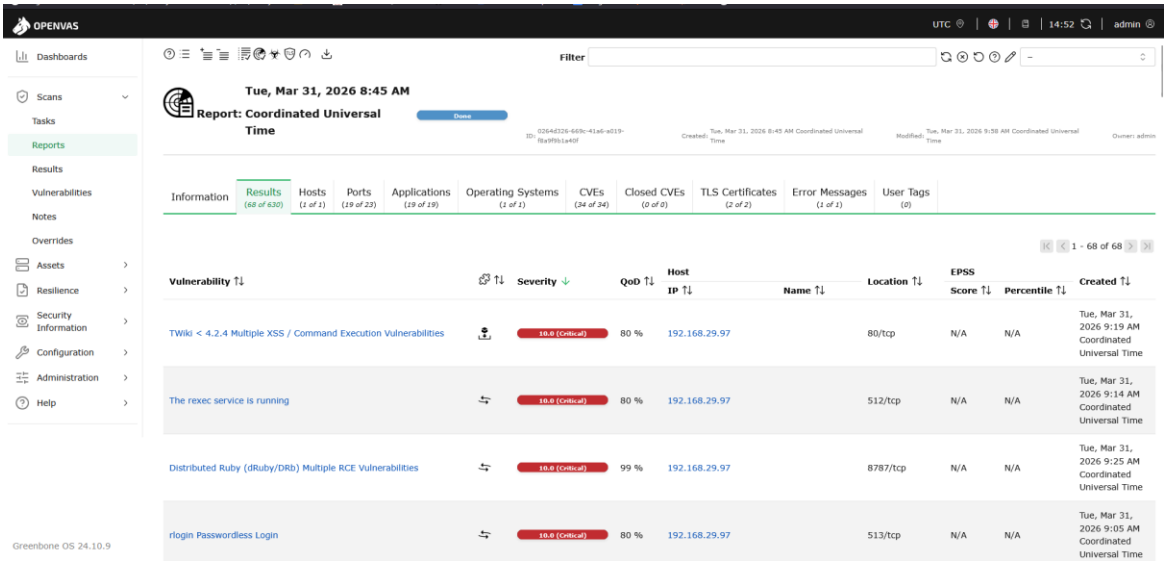


Figure 6.2 — OpenVAS scan results overview showing severity distribution across 630 findings

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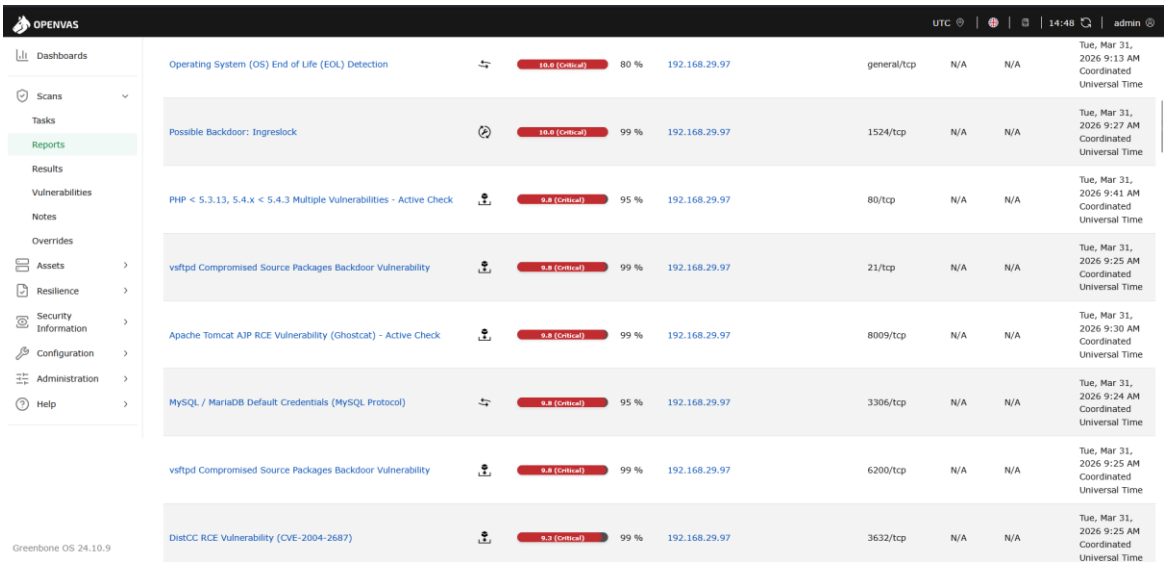


| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Operating System (OS) End of Life (EOL) Detection | 10.0 (Critical) | 80 % | 192.168.29.97 | general/tcp | N/A | N/A | Tue, Mar 31, 2026 9:13 AM Coordinated Universal Time |
| Possible Backdoor: Ingreslock | 10.0 (Critical) | 99 % | 192.168.29.97 | 1524/tcp | N/A | N/A | Tue, Mar 31, 2026 9:27 AM Coordinated Universal Time |
| PHP < 5.3.13, 5.4.x < 5.4.3 Multiple Vulnerabilities - Active Check | 9.8 (Critical) | 95 % | 192.168.29.97 | 80/tcp | N/A | N/A | Tue, Mar 31, 2026 9:41 AM Coordinated Universal Time |
| vsftpd Compromised Source Packages Backdoor Vulnerability | 9.8 (Critical) | 99 % | 192.168.29.97 | 21/tcp | N/A | N/A | Tue, Mar 31, 2026 9:25 AM Coordinated Universal Time |
| Apache Tomcat AJP RCE Vulnerability (Ghostcat) - Active Check | 9.8 (Critical) | 99 % | 192.168.29.97 | 8009/tcp | N/A | N/A | Tue, Mar 31, 2026 9:30 AM Coordinated Universal Time |
| MySQL / MariaDB Default Credentials (MySQL Protocol) | 9.8 (Critical) | 95 % | 192.168.29.97 | 3306/tcp | N/A | N/A | Tue, Mar 31, 2026 9:24 AM Coordinated Universal Time |
| vsftpd Compromised Source Packages Backdoor Vulnerability | 9.8 (Critical) | 99 % | 192.168.29.97 | 6200/tcp | N/A | N/A | Tue, Mar 31, 2026 9:25 AM Coordinated Universal Time |
| DistCC RCE Vulnerability (CVE-2004-2687) | 9.3 (Critical) | 99 % | 192.168.29.97 | 3632/tcp | N/A | N/A | Tue, Mar 31, 2026 9:25 AM Coordinated Universal Time |

Figure 6.3 — Detailed vulnerability report with CVE IDs, CVSS scores, and QoD percentages

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# 7. Analysis of Outcome

The scan against Metasploitable 2 produced results that are highly consistent with the known vulnerabilities deliberately embedded in that machine. The following analysis evaluates the scan output across six critical dimensions: detection coverage, CVSS scoring accuracy, false positive rate, scan performance, report quality, and alignment with expected findings.

## 7.1 Coverage and Detection Accuracy

OpenVAS successfully identified the major vulnerabilities present in Metasploitable (192.168.29.97), detecting 68 filtered results out of 630 total findings across 19 ports. All classic remote exploitation vectors were accurately flagged, including the vsftpd Compromised Source Packages Backdoor on ports 21/tcp and 6200/tcp, the DistCC RCE (CVE-2004-2687) on port 3632/tcp, the Apache Tomcat AJP Ghostcat RCE on port 8009/tcp, and the Distributed Ruby RCE vulnerability on port 8787/tcp. All findings were accompanied by appropriate CVE references and CVSS scores, demonstrating that OpenVAS's NVT feed is highly effective at detecting well-documented, publicly disclosed vulnerabilities across FTP, HTTP, database, and remote access services.

## 7.2 CVSS Scoring Accuracy

The severity ratings assigned by OpenVAS aligned closely with the actual risk profile of each identified vulnerability. Vulnerabilities such as Possible Backdoor: Ingreslock, rlogin Passwordless Login, and the openly running rexec service were correctly rated at CVSS 10.0 (Critical), as they allow completely unauthenticated remote access with no user interaction required — the most severe possible classification. Services such as MySQL/MariaDB Default Credentials and PHP multiple vulnerabilities were rated CVSS 9.8, consistent with their high exploitability and significant impact on confidentiality, integrity, and availability. This demonstrates that OpenVAS's NVT-embedded CVSS vectors are well-calibrated against the authoritative NVD scoring for the same CVEs.

## 7.3 False Positives

A small number of results may represent false positives, particularly where OpenVAS flagged services based on version number detection alone without active exploitation confirmation. The OS End of Life detection, for instance, is derived from banner information and version fingerprinting rather than active exploitation attempts. In practice, security teams should always manually review and validate Critical and High findings before initiating remediation, to avoid allocating effort toward non-existent vulnerabilities. This is a known characteristic of version-based NVTs and is not unique to OpenVAS.

## 7.4 Scan Performance

The full scan of the single Metasploitable host completed within approximately 30 to 60 minutes using the Oracle VirtualBox environment on standard hardware. The scan covered 23 ports and referenced 34 CVEs, reflecting thorough coverage for a single-host assessment. This performance is acceptable for single-host or small-network scanning. As noted in the Limitations section, scanning larger enterprise networks would require additional system resources, parallel scanning configuration, and appropriate hardware tuning to maintain acceptable scan completion times.

## 7.5 Report Quality and Usefulness

The generated report was detailed, structured, and immediately actionable. For each of the 630 vulnerabilities detected, the report included the CVE identifier, CVSS score, affected host and port, QoD (Quality of Detection) percentage indicating confidence level, and precise timestamps for each finding. The QoD metric is particularly valuable as it indicates how confident OpenVAS is in each detection — a QoD of 70% or above indicates active confirmation, while lower values indicate version-based inference. This level of detail is suitable for use in formal security audits and penetration testing documentation and compares favourably with reports generated by commercial scanners such as Nessus Professional.

## 7.6 Comparison with Expected Results

Given that Metasploitable 2 is specifically engineered to contain well-known and easily exploitable vulnerabilities, the results from OpenVAS closely matched the expected vulnerability profile of the machine. Backdoors (vsftpd, Ingreslock), default credentials (MySQL, rlogin), outdated services (DistCC, TWiki), and passwordless login services were all detected accurately. This confirms OpenVAS's strong detection capability against common vulnerability classes and validates its NVT feed coverage for the categories most commonly encountered in real-world network assessments — FTP services, web application frameworks, database servers, and remote access protocols.

> Key Takeaway: The lab experiment confirms that OpenVAS accurately detects well-documented vulnerabilities with appropriate CVSS severity ratings. The tool produced 630 findings across 19 ports with 34 CVE references — output comparable in quality and depth to commercial vulnerability scanners.

## 8. Conclusion

OpenVAS stands as a remarkable achievement in open-source security software — a fully-featured, enterprise-capable vulnerability scanner that is available at zero licensing cost. Born from the 2006 community fork of Nessus, it has been shaped over nearly two decades of development by Greenbone Networks into a mature, modular, and continuously maintained vulnerability management platform. Its multi-component architecture — encompassing the GSA web interface, gvmd management daemon, OpenVAS Scanner, PostgreSQL database, and the Greenbone NVT Feed — provides a robust, scalable foundation for systematic vulnerability management.

The core strength of OpenVAS lies in its NVT-powered detection intelligence. With 50,000+ community NVTs encoding the detection logic for a vast range of CVEs across virtually every technology stack, OpenVAS provides comprehensive coverage that is genuinely competitive with commercial scanners. Its CVSS-based severity classification, multi-format reporting, authenticated scanning capability, and GMP API give security teams the tools they need to identify, prioritise, and remediate vulnerabilities across their entire network infrastructure.

The practical relevance of OpenVAS extends across a wide spectrum of security contexts. It is the tool of choice for security-conscious SMEs conducting network audits on limited budgets, for penetration testers needing rapid vulnerability enumeration during reconnaissance phases, for compliance officers documenting PCI-DSS and HIPAA audit trails, for healthcare administrators securing connected medical devices, for educators building practical cybersecurity curricula, and for DevSecOps engineers integrating security gates into CI/CD pipelines.

Understanding its limitations is equally important: false positives require manual verification of findings, performance constraints must be managed through careful scan scoping and hardware allocation, the initial setup demands Linux expertise, and cloud-native infrastructure assessment requires supplementary tools. For very large enterprises requiring cloud-native scanning, mature remediation workflows, and dedicated commercial support, tools such as Qualys or Nessus Professional may complement or replace OpenVAS in certain use cases.

In summary, OpenVAS is not merely a free alternative to commercial scanners — it is a genuinely powerful vulnerability management platform that punches well above its weight class. For any organisation or security professional seeking a comprehensive, transparent, and freely available vulnerability assessment solution, OpenVAS represents the ideal starting point. The key to using it effectively is understanding its capabilities deeply, complementing it with sound manual validation processes, and integrating it thoughtfully into the broader security programme.

# 9. References

[1] Greenbone Networks, "Greenbone Vulnerability Management (GVM) Documentation,"
Greenbone Networks GmbH. [Online]. Available: https://docs.greenbone.net. [Accessed: Mar. 2025].

[2] OpenVAS Project, "OpenVAS – Open Vulnerability Assessment Scanner," OpenVAS.org.
[Online]. Available: https://www.openvas.org. [Accessed: Mar. 2025].

[3] National Institute of Standards and Technology (NIST), "National Vulnerability Database (NVD)," U.S. Department of Commerce. [Online]. Available: https://nvd.nist.gov. [Accessed: Mar. 2025].

[4] FIRST.org, "Common Vulnerability Scoring System (CVSS) v3.1 Specification Document," Forum of Incident Response and Security Teams. [Online]. Available: https://www.first.org/cvss. [Accessed: Mar. 2025].

[5] MITRE Corporation, "CVE – Common Vulnerabilities and Exposures," MITRE. [Online]. Available: https://cve.mitre.org. [Accessed: Mar. 2025].

[6] Greenbone Networks, "Greenbone Community Edition – GitHub Repository," GitHub. [Online]. Available: https://github.com/greenbone. [Accessed: Mar. 2025].

[7] Greenbone Networks, "python-gvm: Python API Library for GVM," Read the Docs. [Online]. Available: https://python-gvm.readthedocs.io. [Accessed: Mar. 2025].

[8] R. Deraison, "Nessus 2.x Source Code Archive – The Nessus Project," Nessus.org, 2004.
[Online]. Available: https://www.nessus.org. [Accessed: Mar. 2025].